

Curriculum Vitae

Devon J. Boland, Ph.D.

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Texas A&M University – Division of Research
Whole System Genome Institute
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Education:

Ph.D.	Biochemistry & Molecular Biophysics, 2023 Texas A&M University, College Station, TX.
B.S.	Chemistry/Biochemistry, 2018 Northern Illinois University, Dekalb, IL.

Professional Affiliations:

2024-present	Board of Directors, Texas Genetics Society
2024-present	Core Facility Showcase Committee Member, Texas A&M University
2020-2023	American Society of Plant Biologists
2019-2020	Graduate Senate Representative, Texas A&M University
2015-2018	American Chemical Society
2014-present	Alpha Chi Sigma

Positions Held:

2025-present	Graduate Course Instructor, Biochemistry and Biophysics Department, Texas A&M University
2023-present	Assistant Research Scientist TIGSS Bioinformatics Core, Texas A&M University
2023-present	Certification in Bioinformatics & Computational Biology (Cert-BICB) Instructor, Texas A&M University
2022-2023	AgriLife Graduate Research Scholar, Texas A&M University
2018-2023	Teaching Assistant, Texas A&M University
2017-2018	Undergraduate Research Assistant, NIGMS R15, Northern Illinois University

Professional Experience:

2024	Fullbright Review Committee, Texas A&M University
2024-present	Website Coordinator, Texas Genetics Society
2024-present	Annual Meeting Sponsorship Coordinator, Texas Genetics Society
2024-present	External Collaborator, Behavioral Plasticity Research Institute
2024-present	Guest Lecturer for BICH 414, Texas A&M University
2024	Guest Lecturer for BICH 464, Texas A&M University
2023	Data Parallelism Workshop, NVIDIA/Texas A&M University
2022-present	HPRC Workshop Leader, Texas A&M University

Teaching Experience:

2025-present	GENE 658: Differential Gene Expression, Instructor Genetics and Genomics Interdisciplinary Graduate Program Texas A&M University
2025-present	BICH 689: Introduction to Protein Structure Prediction, Instructor Biochemistry & Biophysics Department, Texas A&M University
2023-2024	GENE 689: Introduction to AlphaFold2, Instructor Genetics and Genomics Interdisciplinary Graduate Program Texas A&M University
2018-2023	BICH 414: Biochemical Techniques I, Teaching Assistant Biochemistry & Biophysics Department, Texas A&M University
2018-2023	BICH 412: Biochemistry Laboratory I, Teaching Assistant Biochemistry & Biophysics Department, Texas A&M University

Awards & Honors:

2025	Distinguished Service Award, Texas Genetics Society
2023	2 nd Place Oral Presentation Award, Texas A&M University
2022-2023	Norman Borlaug Endowed Research Scholar, Texas A&M University
2022	2 nd Place Award Life Sciences Presentation, Texas A&M University
2022	Paul Barry Gayle Teaching Award, Texas A&M University
2017	Ed Hyland Scholarship Award, Northern Illinois University

Grants & Funding:

Research Grant 2/18/25
National Science Foundation \$15,000,000
Status – Under Review
P.I. Hojun Song
Role: Early Career Scientist/Senior Personnel
“BII: BPRI 2.0 – Accelerating Behavioral Plasticity Research Through Scalable Interdisciplinary Team Science”

S10 Instrumentation Grant 6/03/24
National Institute of Health \$831,510
Status – Not Accepted
P.I. Chendil Damodaran
Role: Bioinformatician/Senior Personnel
“Hyperion Imaging Mass Cytometry”

Research Grant 2/21/23
National Science Foundation \$1,305,453
Status – Not Accepted
P.I. Timothy P. Devarenne
Role: Graduate Student Research Assistant
“RESEARCH-PGR: Deciphering hydrocarbon and polysaccharide production in the green microalga *Botryococcus braunii*.”

Fellowship
National Science Foundation
Status – Not Accepted
P.I. Devon J. Boland
Role: Submitter
“Graduate Research Fellowship Program”

Articles:

1. Yu, L., Yan, M., Knue, J.D., Smith, H.M., Wolff, S.M., Dolan, C.P., Muneoka, K., Romero, S., Cai, J., Yun, C., **Boland, D.J.**, Brunauer, R., Dawson, L.A. (2025) *In Revision*. FGF8 induced bone and joint regeneration at middle phalanx amputation wounds in neonate mice. NPJ Regenerative Medicine.
2. Mohajeri, A., Shin, S. Y., Padgham, S., **Boland, D. J.**, Ratterree, D. P., Blizman, J., Han, G., Woodman, C., Trache, A. (2025) *In Review*. Mechanical signaling regulates vascular smooth muscle cells adaptation in aged resistance arteries. Frontiers in Physiology.
3. Novikov, B., **Boland, D. J.**, Metsalov, I., Scott, H., Dauletbayeva, S., Monagas-Valentin, D., Panin, V. (2025) *In Revision*. CMP-sialic acid synthetase in *Drosophila* requires N-1 glycosylation of a non-canonical site. Journal of Biological Chemistry.
4. Ogunlusi, O., Sarkar, M., Chakrabarti, A., **Boland, D. J.**, Nguyen, T., Sampson, J., Nguyen, C., Fails, D., Jones-Hall, Y., Fu, L., Mallick, B., Keene, A., Jones, J., & Sarkar, T. R. (2024). Circadian rhythm disruption alters mammary gland morphology and accelerates “cold” aggressive tumorigenesis through a LILRB4-dependent pathway. bioRxiv, 2024.2003.2019.585534. <https://doi.org/10.1101/2024.03.19.585534>
5. **Boland, D. J.**, Cornejo-Corona, I., Browne, D. R., Murphy, R. L., Mullet, J., Okada, S., & Devarenne, T. P. (2024). Reclassification of *Botryococcus braunii* chemical races into separate species based on a comparative genomics analysis. PLOS ONE, 19(7), e0304144. <https://doi.org/10.1371/journal.pone.0304144>
6. **Boland, D. J.**, & Ayres, N. M. (2024). Cracking AlphaFold2: Leveraging the power of artificial intelligence in undergraduate biochemistry curriculums. PLOS Computational Biology, 20(6), e1012123. <https://doi.org/10.1371/journal.pcbi.1012123>
7. Cornejo-Corona, I., **Boland, D. J.**, & Devarenne, T. P. (2024). Method for isolation of high molecular weight genomic DNA from *Botryococcus* biomass. PLOS ONE, 19(7), e0301680. <https://doi.org/10.1371/journal.pone.0301680>
8. **Boland, D. J.** (2023). AN INVESTIGATION OF THE GENOMIC SPACE AND SEARCH FOR HYDROCARBON BIOSYNTHETIC GENES IN THE GREEN MICROALGA BOTRYOCOCCUS BRAUNII. *TAMU Thesis*.
9. Amaro, N. A., Ammerman, K. R., **Boland, D. J.**, Bork, C. J., Davis, J. V., Haskell, D. W., Hoeksema, C., Juskevics, S. J., Obrycki, K. M., Pacheco, G., Padilla, N., Rasmussen, R. A., Riggins-Walker, T. R., Sattar, Z. S., Skrypai, Y., Hoerchler, K. B., De Lio, A. M., & Gilbert, T. M. (2019). Computational Studies of Relative Stabilities of Low-Spin d(6) cis- and trans-[M(en)2X2](+) Complexes

(M = Co, Rh, Ir): Steric and Electronic Effects in the Context of the Structural Trans Influence. *J Phys Chem A*, 123(12), 2438-2446. <https://doi.org/10.1021/acs.jpca.9b01740>

Invited Talks & Presentations:

1. **Devon J. Boland**, and Charles M. Dickens. Introduction to AlphaFold2 on Grace. 5th Annual Texas A&M University Research Computing Symposium. Local symposium. Guest speaker/workshop leader, May 23rd-25th, 2022. Texas A&M University.
2. **Devon J. Boland**, and Timothy P. Devarenne. Looking Beyond Hydrocarbon Production in *Botryococcus braunii*. 2021 Genome Editing Symposium: Air, Land, and Sea. Local symposium, poster, October 27-29th, 2021. Texas A&M University.
3. **Devon J. Boland**, and Timothy P. Devarenne. Comparative Genomic Analysis of *Botryococcus braunii*. 1st International Symposium on Omic Technologies. International meeting, talk, October 20-22nd, 2021. Universidad De Antioquia, Columbia.
4. **Devon J. Boland**, Dionne H. Griffin, Christopher Smith, and James R. Horn. Engineering a pH-Dependent Anti- Hapten VHH Antibody. 30th Annual Gibbs Conference on Bio-thermodynamics. National meeting, poster, September 24-27th, 2016. Carbondale, IL.
5. **Devon J. Boland**, Emily E. Fiala, Dionne H. Griffin, and James R. Horn. Engineering a pH-Dependent Molecular Switch for Anti-Methotrexate VHH. Summer Research Symposium. University conference, talk, August 12th, 2016. Northern Illinois University.
6. **Devon J. Boland**, Dionne H. Griffin, and James R. Horn. Evaluation of a Module Based, Bivalent Reagent for Immunoassays. Undergraduate Research and Artistry Day 2016. University conference, poster, April 19th, 2016, Northern Illinois University.
7. **Devon J. Boland**, Dionne H. Griffin, and James R. Horn. Engineering a Module Based, Bivalent Reagent for Immunoassays. Undergraduate Research and Artistry Day 2015. University conference, poster, April 17th, 2015, Northern Illinois University.

Mentoring Experience:

2025-present	Carrisa Yun M.S., Internship Mentor
2024-present	Dongmei Zhang Ph.D., Joint Mentorship with Dr. Jeff Cirillo MPIM.
2022-2023	Gabriele Golen, Undergraduate Researcher
2021-2022	Cecelia Hambleton, Undergraduate Researcher
2017-2018	Victoria Pho, Undergraduate Researcher
2016-2017	Emily Fiala, Undergraduate Researcher